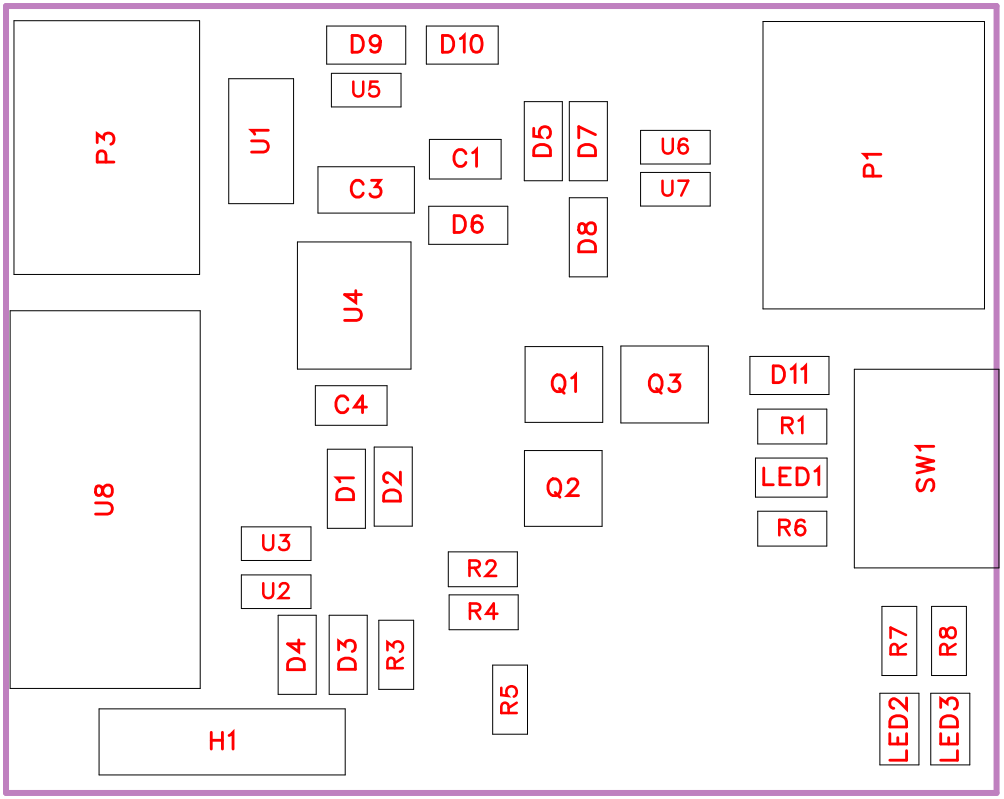
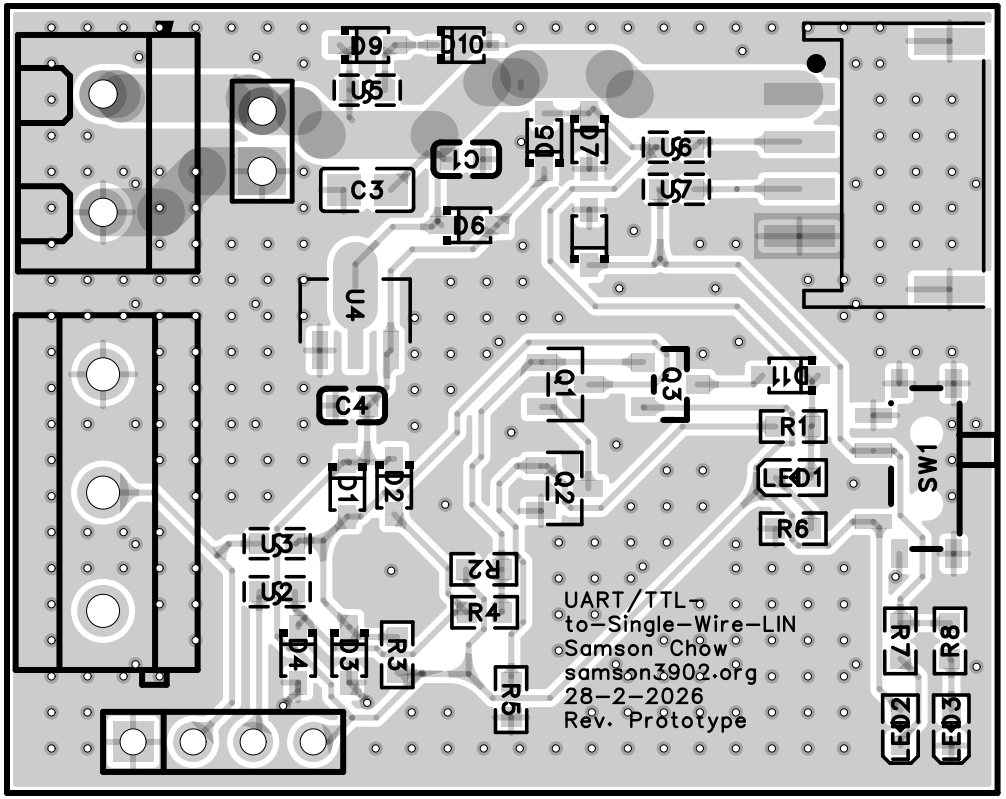
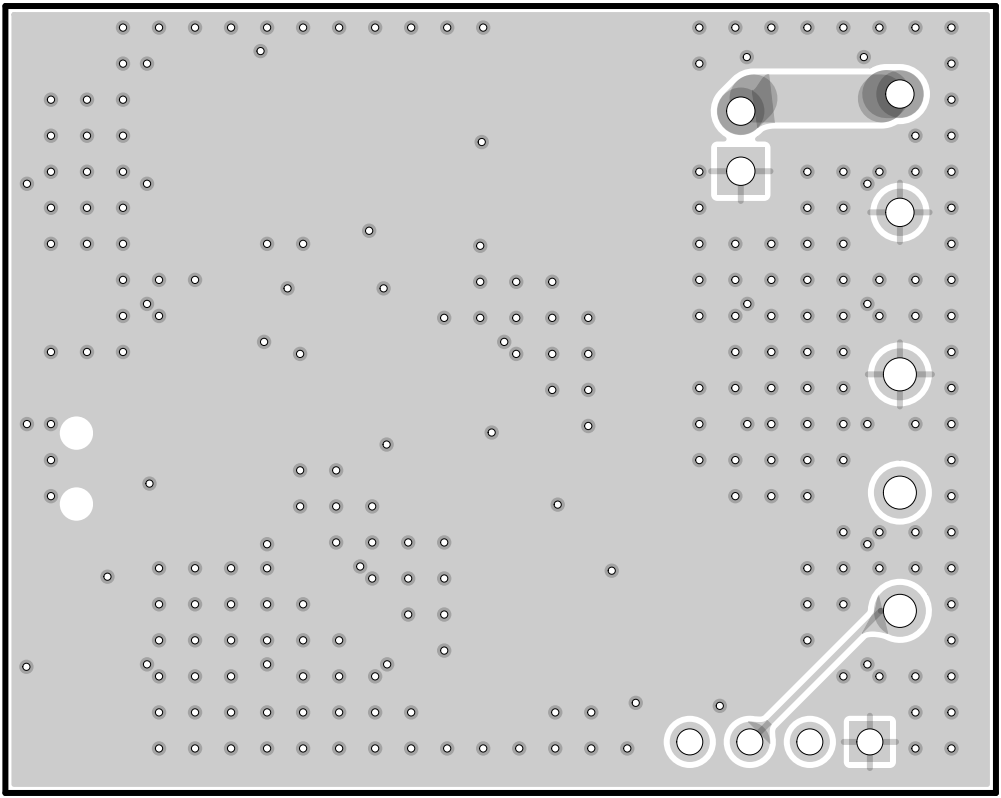
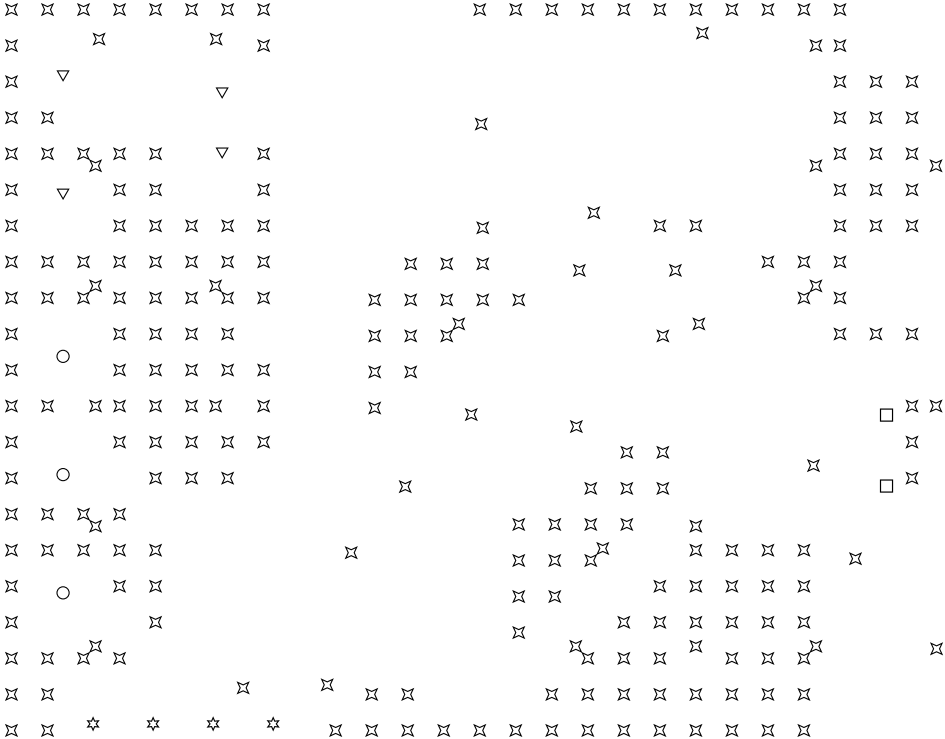

P3
U1
D9
D10
U5
C1
C3
D6
D5
D7
D8
U6
U7
P1
U4
U8
C4
Q1
Q3
Q2
D11
R1
LED1
R6
SW1
D1
D2
U3
U2
R2
R4
D4
D3
R3
R5
R7
R8
LED2
LED3
H1

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	100nF	C1	C0603	100nF	CGA0603X7R104K500JT	HRE(芯声)	C6119867	LCSC
2	1	10uF	C3	C0805	10uF	CGA0805X5R106K500MT	HRE(芯声)	C6119889	LCSC
3	1	1uF	C4	C0603	1uF	CGA0603X5R105K500JT	HRE(芯声)	C6119852	LCSC
4	3	1N4148WS	D1,D2,D11	SOD-323_L1.8-W1.3-LS2.5-RD		1N4148WS	DIODES(美台)	C526200	LCSC
5	4	SD05	D3,D4,D7,D8	SOD-323_L1.8-W1.3-LS2.5-RD		SD05		C127518	LCSC
6	3	1N4148WS-7-F-79	D5,D6,D9	SOD-323_L1.8-W1.3-LS2.5-RD		1N4148WS-7-F-79	DIODES(美台)	C526200	LCSC
7	1	SD36	D10	SOD-323_L1.7-W1.3-LS2.7-RD		SD36	MDD(辰达半导体)	C502531	LCSC
8	1	HDR2.54-LI-2x4P	H1	HDR-TH_4P-P2.54-V-M		HDR2.54-LI-2x4P		C9900020145	LCSC
9	3	KT-0603R	LED1,LED2,LED3	LED0603-RD		KT-0603R	KENTO	C2286	LCSC
10	1	PH2.0-4PWT	P1	CONN-SMD_PH2.0-4PWB		PH2.0-4PWT		C9900002982	LCSC
11	1	KF301-2P接线端子	P3	CONN-TH_XY301V-A-5.0-2P		KF301-2P接线端子		C9900016950	LCSC
12	2	2N7002	Q1,Q2	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		2N7002	CJ(江苏长电/长晶)	C8545	LCSC
13	1	BSS84	Q3	SOT-23_L2.9-W1.3-P1.90-LS2.4-BR		BSS84	FUXINSEMI(富芯森美)	C3018484	LCSC
14	5	1kΩ	R1,R4,R6,R7,R8	R0603	1kΩ	0603WAF1001T5E	UNI-ROYAL(厚声)	C21190	LCSC
15	3	4.7kΩ	R2,R3,R5	R0603	4.7kΩ	0603WAF4701T5E	UNI-ROYAL(厚声)	C23162	LCSC
16	1	MSK12C02	SW1	SW-TH_MSK12C02		MSK12C02	SHOU HAN(首韩)	C431540	LCSC
17	1	HEAD2	U1	HDR-TH_2P-P2.54-V-M		HEAD2		C9900010030	LCSC
18	5	SMD0603-002/60N	U2,U3,U5,U6,U7	F0603		SMD0603-002/60N	BORN(伯恩半导体)	C2687869	LCSC
19	1	HT7550-1-36V	U4	SOT-89-3_L4.3-W2.3-P1.50-LS4.3-BR		HT7550-1-36V	TECH PUBLIC(台舟)	C53340891	LCSC

20	1	KF301-5.0-3P	U8	CONN-TH_3P-P5.00_KF301-5.0-3P		KF301-5.0-3P	KEFA(科发)	C474882	LCSC


D9
D10
U5
C1
D5
D7
U6
U7
C3
D6
U4
Q1
Q3
D11
C4
R1
LED1
SW1
D1
D2
Q2
R6
U3
U2
R2
R4
UART/TTL-
to-Single-Wire-LIN
Samson Chow
samson3902.org
28-2-2026
Rev. Prototype
D4
D3
R3
R5
R7
R8
LED2
LED3